Collaborative Context-Affine Perception Network for Remote Sensing Small Object Detection

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***Abstract*—Small object detection in remote sensing images (RSIs) is challenging because imaging degradation weakens object textures, reduces contrast, and blurs boundaries. These effects are further aggravated by hierarchical feature extraction, where repeated downsampling weakens shallow spatial cues before they reach deeper semantic representations. To alleviate these effects, we propose the collaborative context-affine perception network (CoCAPNet), which progressively refines structural, contextual, and spatial representations along the detection pipeline. CoCAPNet comprises four complementary components. First, the context-aware dynamic affine module (CDAM) separates feature responses into structure- and detail-dominant components and adaptively recombines them to retain weak spatial cues. Next, the dual-range perception fusion module (DPFM) adaptively balances local detail and global semantics by integrating scale-dependent response predictions with multilevel receptive field processing. The collaborative multikernel block (CMKB) then addresses sparse and unstable activations in small objects by enhancing object-relevant patterns via dynamic multikernel perception, fine-grained refinement, and channel-aware weighting. Finally, the fine-grained small object detector (FSDetector) introduces a high-resolution prediction branch to retain spatial detail for small targets. These components form a progressive feature-processing pipeline from representation reconstruction to scale-aware fusion, structural refinement, and high-resolution prediction. Experiments on the DIOR, NWPU VHR-10, and AI-TOD yield mean average precision (mAP) values of 86.3%, 95.8%, and 59.1%, respectively, indicating consistent gains across datasets with substantial scale variation and densely distributed small objects.**

***Index Terms*—Context-aware feature integration, multiscale affine perception, remote sensing image (RSI), small object detection.**

I. Introduction

With the rapid advancement of satellite remote sensing technologies, object detection in remote sensing imagery has become a fundamental component in aerospace [1], disaster monitoring [2], and environmental assessment [3]. However, remote sensing images (RSIs) are affected by sensor characteristics, atmospheric conditions, spatial resolution, and the modulation transfer function (MTF), all of which can attenuate fine-scale image information. For small objects, these effects manifest as weak textures, low contrast, and poorly defined boundaries.

These degradations can be further aggravated during hierarchical feature extraction, as repeated downsampling progressively reduces spatial detail and weakens the separability of small objects from their surroundings. Consequently, small targets may become difficult to separate from background structures in deeper feature spaces, as illustrated in Fig. 1. This semantic degradation, induced by imaging deterioration, constitutes a critical bottleneck in small object detection within RSIs.

To counteract degradation-induced losses, recent studies have introduced specialized detection frameworks for RSIs, all aiming to strengthen small object representation under complex imaging conditions. Li et al. [4], for example, combine attention-guided sampling with feature magnification to emphasize difficult regions. CLT-Det [5] uses feature

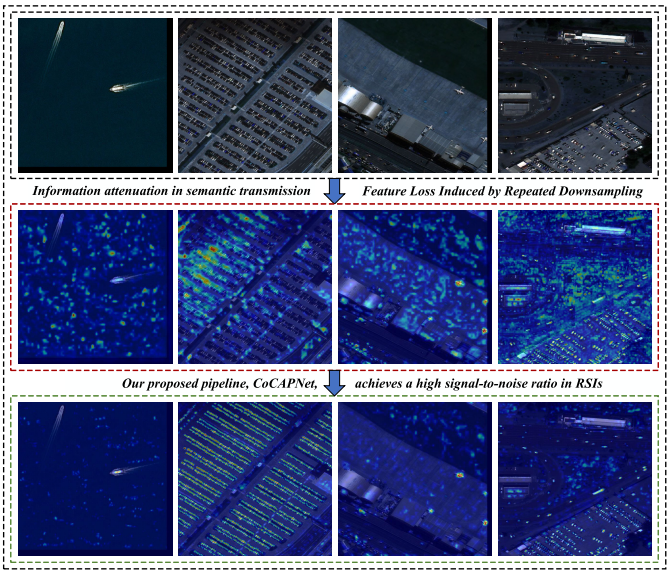


Fig. 1. An illustration of small object feature degradation and its mitigation by CoCAPNet. The progression from the original image to the baseline's weak response, and finally to our method's distinct heatmap, illustrates the stronger object responses obtained with CoCAPNet compared with the baseline.

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refinement to reduce scale-related semantic inconsistency, while Position-DETR [6] progressively selects foreground regions and refines attention. Huo et al. [7] exploit long-range context through a global guidance branch. Although these methods improve feature representation, they generally address feature degradation and hierarchical downsampling as separate issues. In particular, weak high-frequency content at shallow stages can be progressively diluted, allowing small object responses to be overwhelmed by background activations during deeper feature fusion.

To address this progressive loss, we propose the collaborative context-affine perception network (CoCAPNet). Rather than treating weak texture, scale variation, and spatial detail loss as isolated problems, CoCAPNet organizes them along the feature-processing path. Context-aware dynamic affine module (CDAM) preserves complementary structural and detail cues; the dual-range perception fusion module (DPFM) adapts the receptive-field contribution to scale; collaborative multikernel block (CMKB) refines sparse and heterogeneous responses; and the FSDetector retains high-resolution information at the prediction stage.

CoCAPNet contains four complementary components: the CDAM, DPFM, CMKB, and fine-grained small object detector (FSDetector). CDAM reconstructs structure- and detail-dominated responses during feature propagation. DPFM performs adaptive multiscale fusion between local and global context. CMKB refines the resulting representations through multikernel perception, local processing, and channel reweighting. The FSDetector then exploits a high-resolution prediction level to retain spatial detail during detection.

These components define a progressive feature-processing strategy from representation reconstruction to scale-aware fusion, structural refinement, and high-resolution prediction. The contributions of this research are as follows.

1) We propose CoCAPNet, a unified small object detector that explicitly addresses the progressive attenuation of weak object cues during hierarchical feature extraction and prediction.
2) We design the‘ CDAM to reconstruct structure- and detail-dominated responses in the spatial domain. Its adaptive fusion preserves fine-grained cues while incorporating contextual information, providing a more stable representation for weak-textured targets.
3) We develop the DPFM for scale-aware feature fusion. Its predicted scale weights and dual-range processing combine local detail with broader context without relying on a single fixed receptive field.
4) We introduce CMKB and the FSDetector as complementary refinement and prediction mechanisms. CMKB improves the stability of sparse object responses through multikernel and channel-aware processing, while FSDetector exploits a high-resolution prediction level to improve localization of small targets.

The remainder of this article is organized as follows. Section II reviews related work. Section III describes the proposed method in detail. Section IV presents experimental results and comparisons with state-of-the-art approaches. Section V concludes this article.

II. Related Work

A. Imagery Degradation and Semantic Enhancement

RSI quality is affected by spatial resolution, sensor characteristics, atmospheric conditions, and the MTF of the imaging system. The resulting attenuation of high-frequency content weakens boundaries and texture, which is particularly detrimental to objects represented by only a small number of pixels. When such features enter a hierarchical detector, subsequent downsampling can further reduce their separability from surrounding structures.

Early approaches addressed image degradation at the pixel level using contrast enhancement, histogram equalization, or Retinex-based correction. Such preprocessing can improve visual quality but does not adapt to the evolving representation within a detector. However, these pixel-domain methods cannot adaptively adjust feature semantics within deep models. Deep restoration and superresolution networks provide a more learned alternative, yet their additional computation and susceptibility to reconstruction artifacts can limit their suitability for detection pipelines.

Attention mechanisms provide another route by recalibrating feature responses. SENet [8] performs channel reweighting via global average pooling, while CBAM [9] and ECA [10] incorporate spatial and channelwise calibration. Transformer-based global attention further captures long-range dependencies, and these techniques have been extended to remote sensing to emphasize salient regions and suppress background noise. However, most of these methods focus on global context or channel weighting and overlook the semantic degradation arising from weak textures and blurred edges in shallow layers.

To address these limitations, recent works highlight context-aware feature aggregation. Jiang et al. [11] inferred textures and salient edges to enhance contours for improved super-resolution. Zhang et al. [12] introduced FFCA-YOLO with lightweight modules for better spatial context under real-time constraints. Ming et al. [13] proposed CFC-Net to reduce classification–localization misalignment, while Zhan et al. [14] developed HKFCB to enhance small object responses through BiPDC and heterogeneous kernel fusion. Zhang et al. [15] proposed VTSaRNet, which integrates an instance segmentation-based copy-paste (ISCP) module with offline-online Mosaic augmentation to generate synthetic samples, and employs a bimodal fusion transformer (BFT) to enhance RGB-T person detection in sparse, low-light aerial scenes.

B. Multiscale Perception and Small Object Detection

Scale variation is another central difficulty in RSI detection. Differences in imaging resolution, acquisition altitude, and viewpoint produce substantial variation in object size, while repeated downsampling removes low-level spatial detail. Meanwhile, repeated downsampling in deep convolutional networks causes the loss of low-level details and misalignment of cross-layer semantics, allowing small object responses to be overwhelmed by large-object features during fusion.

To tackle these issues, multiscale feature fusion frameworks have been widely studied. FPN [16] and PANet establish

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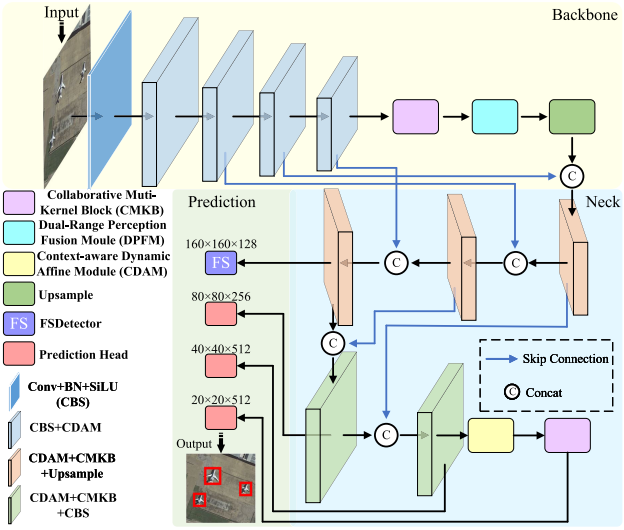


Fig. 2. Overall framework of CoCAPNet. CoCAPNet adopts the anchor-free YOLOv8 detector as its baseline and leverages CSPDarkNet53 as the backbone for multiscale feature extraction from input RSIs.

bidirectional cross-layer pathways for semantic injection, while subsequent works such as NAS-FPN [17], BiFPN [18], and YOLO-series modules further enhance aggregation efficiency. Multibranch perception modules like RFBNet [19] and LSKNet [20] enrich multiscale representations, and spatial pyramid pooling structures [21], [22], [23] improve scale robustness. Recently, adaptive kernel strategies have been investigated for small object scenarios. Zhang et al. [24] proposed MKSNet with multikernel perception and dual attention, while Tao et al. [25] developed cross-dimensional feature fusion for enhanced discrimination. Sun et al. [26] proposed a multidomain, weakly decoupled domain generalization framework for fault diagnosis under unknown operating conditions, highlighting the importance of robust multidomain representation learning for handling distribution variations. Despite these advances, existing methods still struggle to simultaneously balance local detail preservation, global contextual modeling, and adaptive receptive field selection for remote sensing small object detection.

Driven by these limitations, we propose three complementary modules, the CDAM, DPFM, and CMKB, designed to alleviate semantic degradation, balance multiscale perception, and enhance structural features, respectively. Together, they form a progressive feature refinement pipeline for small object detection in RSIs.

III. Proposed Method

A. Overview

The overall architecture of CoCAPNet is shown in Fig. 2. Given an input RSI, multiscale feature maps are first extracted by the CSPDarkNet53 backbone [27]. These features are first processed by the CDAM, which separates structure- and detail-dominant responses to retain spatially discriminative information. Through its context-aware dynamic affine transformation, the CDAM adaptively decomposes features into structure-related and detail-related components, enabling more effective preservation of complementary information during feature propagation. In particular, the CDAM obtains a structure-dominant response by downsampling and upsampling the input feature. This operation suppresses fine spatial variations while retaining relatively coarse structural information. Meanwhile, the difference between the original feature and its reconstructed low-frequency counterpart is regarded as high-frequency detail information, which contains object boundaries, edge responses, texture patterns, and other fine-grained cues. For remote sensing small object detection, these high-frequency details are particularly important because small targets usually occupy only a few pixels and rely heavily on subtle boundary and texture characteristics for discrimination. On the contrary, low-frequency structural information provides essential contextual guidance to distinguish target regions from complex backgrounds. By explicitly separating and reconstructing these two complementary components, the CDAM preserves both semantic context and fine-grained discriminative details, thereby mitigating semantic attenuation and enhancing small object representation. To enhance scale adaptability, DPFM integrates multiscale features by dynamically adjusting its receptive field, enabling flexible response selection under scale variation and effectively addressing dense small object distributions and scale imbalance. The refined features are further processed by the CMKB, which fuses local context perception with cross-channel attention to refine semantic responses and suppress background interference. By emphasizing salient regions, the CMKB enhances feature discrimination and stability, ensuring reliable small object perception in cluttered environments. Finally, the enriched and context-interacted features are passed to the FSDetector, which adopts a multiscale decoding structure for accurate localization and classification. The resulting framework jointly refines structural, contextual, and spatial representations before prediction.

B. Context-Aware Dynamic Affine Module

Small objects in RSIs often exhibit weak textures, low contrast, and blurred boundaries, which cause feature confusion across categories when processed by conventional convolutional neural networks. To better disentangle structural information from fine details while improving contextual perception, we propose the CDAM. The module performs a frequency-inspired affine transformation that decomposes the input feature map into a structure-dominated low-frequency component and a detail-dominated high-frequency component. Unlike conventional Fourier-based methods, our approach operates entirely in the spatial domain, reducing the additional computational burden associated with frequency-domain transformation and avoiding potential reconstruction distortions caused by frequency-domain processing.

CDAM consists of three hierarchical submodules: the structure-detail refinement module (SDRM), the context-adaptive enhancement block (CAEB), and the hierarchical bottleneck integrator (HBI). The overall architecture is shown in Fig. 3.

1) Structure-Detail Refinement Module: The SDRM separates the input feature into structure- and detail-dominant

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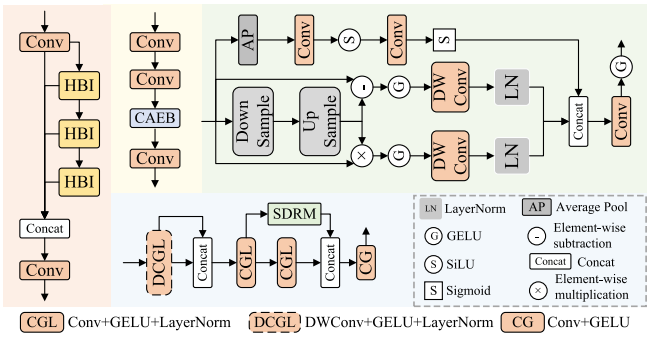


Fig. 3. Overall framework of CDAM. CDAM consists of three synergistic components.

components. Given an input feature map $X \in \mathbb{R}^{B\times C\times H\times W}$, the structure-dominated low-frequency component is extracted via downsampling and upsampling, followed by elementwise modulation

$$X_L = \mathrm{LN}\left(\mathrm{Conv}_L\left(\sigma\left(X \odot \mathrm{Upsample}\left(\mathrm{Down}_k\left(X\right)\right)\right)\right)\right) \tag{1}$$

where $\odot$ denotes elementwise multiplication, $\sigma(\cdot)$ is the GELU activation function, and LN represents layer normalization. The high-frequency detail component is computed as the residual between the original feature and its low-frequency counterpart:

$$X_H = \mathrm{LN}\left(\mathrm{Conv}_H\left(\sigma\left(X - \mathrm{Upsample}\left(\mathrm{Down}_k\left(X\right)\right)\right)\right)\right). \tag{2}$$

To achieve dynamic frequency-aware enhancement, adaptive weights W_L and W_H are generated via sigmoid-gated convolutions, and the final SDRM feature is obtained through weighted fusion

$$X_{\mathrm{SDRM}} = \sigma\left(W_L \odot X_L + W_H \odot X_H\right) \tag{3}$$

where $W_L = \mathrm{Sigmoid}(\mathrm{Conv}_w(X_L))$ and $W_H = \mathrm{Sigmoid}(\mathrm{Conv}_w(X_H))$. This operation allows the network to adaptively adjust the ratio of structure and detail information according to different scenarios, enhancing the discriminability of weak-textured small objects.

2) Context-Adaptive Enhancement Block: The CAEB further processes the decomposed feature by incorporating broader spatial context. The input X_{SDRM} first undergoes depthwise convolution with residual addition

$$X' = X_{\mathrm{SDRM}} + \mathrm{LN}\left(\sigma\left(\mathrm{DWConv}\left(X_{\mathrm{SDRM}}\right)\right)\right). \tag{4}$$

The feature is then channel-compressed and passed through a context convolution to capture wider spatial dependencies, yielding X_c. Finally, the context and detail features are fused with learnable coefficients

$$X_{\mathrm{CAEB}} = \sigma\left(\beta_c \cdot X_c + \beta_e \cdot X_e\right), \quad \beta_c + \beta_e = 1. \tag{5}$$

The fusion combines contextual and detail-related responses, providing both broader spatial information and local feature cues. In particular, context-aware scaling factors and bias terms are introduced to dynamically adjust feature responses, restoring suppressed activations in low-texture regions while enhancing the contrast between target regions and the background.

3) HBI and Overall CDM Structure: The HBI module integrates multilevel features while maintaining residual connections

$$X_{\mathrm{HBI}} = X_{\mathrm{in}} + \mathrm{CAEB}\left(\mathrm{Conv}_2\left(\mathrm{Conv}_1\left(X_{\mathrm{in}}\right)\right)\right). \tag{6}$$

The residual structure facilitates feature reuse and provides a direct path for gradient propagation. CDAM stacks n HBI modules hierarchically. Let $Y_0 = \mathrm{Conv}_{\mathrm{expand}}(X_{\mathrm{in}})$ and $Y_i = \mathrm{HBI}_i(Y_{i-1})$ for $i = 1, \ldots, n$. The final CDAM output is obtained by concatenating all stage outputs

$$X_{\mathrm{CDAM}} = \mathrm{Conv}_{\mathrm{proj}}\left(\left[Y_0, Y_1, \ldots, Y_n\right]\right). \tag{7}$$

The hierarchical integration, combined with frequency-inspired affine separation and context enhancement, effectively preserves high-frequency details while capturing low-frequency structural and contextual information. Through hierarchical feature integration, the CDAM retains both structure-dominant and detail-dominant responses during feature propagation.

C. Dual-Range Perception Fusion Module

Objects in RSIs often exhibit significant scale variations due to imaging resolution, flight altitude, sensor viewpoints, and complex background scenes. During feature extraction and downsampling, small object features are prone to being overwhelmed or lost. To address this, we propose DPFM, as illustrated in Fig. 4. DPFM incorporates a learnable scale-aware weighting mechanism and a multilevel cross-domain fusion strategy, enabling a dynamic balance between local details and global semantic features.

1) Scale-Prediction Mechanism: Given an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C_1\times H\times W}$, we first reduce channel dimensionality via a 1×1 convolution

$$\mathbf{X}_r = \mathrm{Conv}_{1\times 1}\left(\mathbf{X}\right) \tag{8}$$

which reduces channel redundancy and preserves structure-dominant features. The reduced feature map is then passed through a lightweight convolution to predict scale response maps

$$\mathbf{S}_{\mathrm{logit}} = \mathrm{Conv}_{1\times 1}\left(\sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times 3}\left(\mathbf{X}_r\right)\right)\right)\right) \tag{9}$$

where $\sigma(\cdot)$ the denotes ReLU activation and $\mathbf{S}_{\mathrm{logit}} \in \mathbb{R}^{B\times \mathcal{S}\times H\times W}$ represents the raw responses for $\mathcal{S}$ different scales. The resulting logits are normalized with a softmax function to obtain relative weights across the predefined receptive-field branches

$$\alpha_s = \frac{\exp\left(\left(\mathbf{S}_{\mathrm{logit}}\left(s\right) + b_s\right)\right)}{\sum_{j=1}^{\mathcal{S}} \exp\left(\left(\mathbf{S}_{\mathrm{logit}}\left(j\right) + b_j\right)\right)} \tag{10}$$

where b_s is a scale bias term. A regularization term is introduced to stabilize scale learning

$$\mathcal{L}_{\mathrm{scale}} = \sum_{s=1}^{\mathcal{S}} \left\|\alpha_s - \bar{\alpha}\right\|_2^2, \quad \bar{\alpha} = \frac{1}{\mathcal{S}} \sum_{s=1}^{\mathcal{S}} \alpha_s \tag{11}$$

which discourages the weights from becoming excessively concentrated on a single branch.

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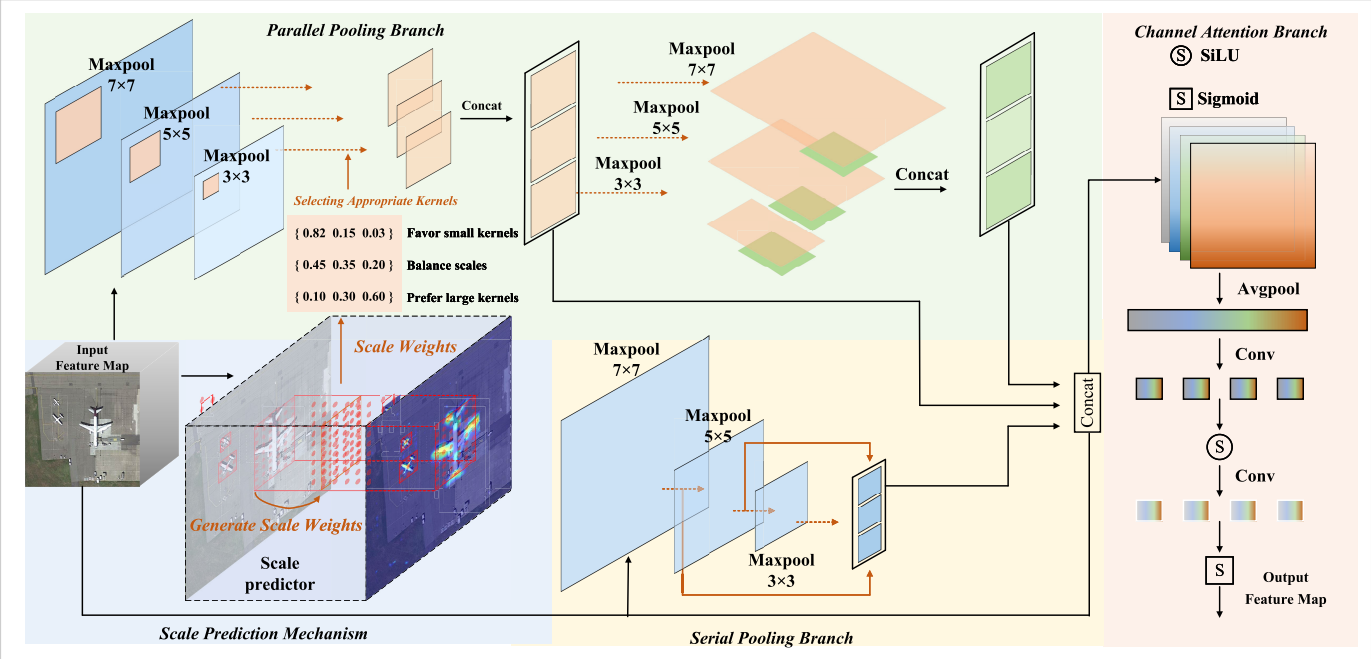


Fig. 4. Structure of DPFM. DPFM comprises a learnable scale-prediction branch and dual-range fusion pathways. The scale-prediction branch generates adaptive receptive-field weights, whereas the local-detail and global-semantic pathways capture fine textures and long-range contextual information, respectively. Through hierarchical fusion, DPFM produces scale-consistent features that retain discriminative cues across diverse object sizes.

2) Multilevel Dual-Domain Fusion: Using the learned scale weights α_s, DPFM extracts multiscale feature responses through pooling operations

$$\mathbf{F}_s = \mathrm{Pool}_s(\mathbf{X}_r), \quad s = 1, 2, \ldots, \mathcal{S}. \tag{12}$$

These features are then aggregated through scale-weighted summation to obtain the first-level dynamic scale fusion feature

$$\mathbf{F}^{(1)} = \sum_{s=1}^{\mathcal{S}} \alpha_s \odot \mathbf{F}_s \tag{13}$$

achieving an *adaptive receptive field reconstruction* that enhances the saliency of small object regions.

A second fusion stage further combines the resulting multiscale responses

$$\mathbf{F}^{(2)} = \sum_{s=1}^{\mathcal{S}} \beta_s \odot \mathrm{Pool}_s\left(\mathbf{F}^{(1)}\right), \quad \beta_s = \mathrm{Softmax}\left(\mathrm{GAP}\left(\mathbf{F}_s\right)\right). \tag{14}$$

The first-level fusion explicitly aggregates multiscale features according to the predicted scale weights α_s, enabling the network to select appropriate receptive fields and emphasize scale-relevant responses for different object sizes. This process mainly focuses on adaptive, scale-aware feature integration. Subsequently, the second-level fusion further models implicit interactions among different scales by applying scale convolution and generating adaptive weights β_s. Unlike the first stage, which performs direct scale selection, this second stage captures cross-scale dependencies and incorporates additional cross-scale contextual information among multiscale representations. Through this two-stage fusion strategy, the DPFM progressively integrates explicit scale preferences and implicit cross-scale interactions, achieving a more balanced representation between local details and global semantics.

In addition, a serial multilayer max pooling path extracts ultralarge receptive field structure features

$$\mathbf{F}^{(3)} = \mathrm{Pool}_{k_{\max}}\left(\mathrm{Pool}_{k_{\max}}\left(\mathrm{Pool}_{k_{\max}}(\mathbf{X}_r)\right)\right) \tag{15}$$

providing an additional large-receptive-field context branch to provide stable global guidance in dense or complex backgrounds.

Finally, all domain features are concatenated and adaptively fused

$$\mathbf{F}_{\mathrm{cat}} = \left[\mathbf{X}_r; \mathbf{F}^{(1)}; \mathbf{F}^{(2)}; \mathbf{F}^{(3)}\right]. \tag{16}$$

The local branch focuses on capturing edge structures and the fine-grained textures of small objects, thereby enhancing spatial discriminability. On the contrary, the global branch enlarges the receptive field to incorporate richer contextual information, thereby improving semantic understanding in complex backgrounds. These two branches form complementary representations.

By dynamically assigning weights to feature responses, the network can adjust the contributions of local and global information according to feature-level scale variations and scene complexity, where the generated weights do not explicitly predict the physical size of objects but represent the relative importance of local and global perception branches in the feature space. This adaptive weighting strategy avoids the bias introduced by single-scale modeling and enables more effective integration of fine-grained details and contextual semantics. Ultimately, the module maintains stable and consistent response distributions across different object scales. It provides balanced and discriminative feature representations

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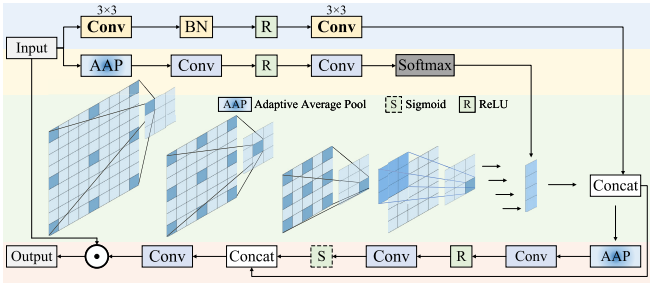


Fig. 5. Structure of the CMKB. The CMKB integrates dynamic kernel selection, local refinement, and channel-aware weighting. It adaptively adjusts receptive fields, restores fine structures, and emphasizes object-relevant cues, producing stable and discriminative features for small object regions.

for subsequent detection heads. The resulting features are then provided to the subsequent detection stages.

D. Collaborative Multikernel Block

In remote sensing imagery, objects exhibit large-scale variations and are often embedded in complex backgrounds. Small objects contain few pixels, resulting in sparse and incomplete feature representations that challenge convolutional structures with fixed receptive fields. These models struggle to balance local detail preservation and global semantic extraction, causing feature degradation and unstable responses. To address this issue, we propose the CMKB, which constructs a collaborative pathway composed of dynamic multikernel selection, local refinement, and channel attention (Fig. 5). Inspired by selective kernel networks [28], CMKB differs from conventional dynamic convolution methods by focusing on feature-level multiscale response enhancement rather than generating or selecting convolution kernels. Through the collaboration of these pathways, the CMKB improves structural adaptivity, restores fine-grained details, and strengthens semantic relevance within the spatial domain.

1) Dynamic Multikernel Selection: Given an input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, a set of dilated depthwise convolutions with different receptive fields captures multiscale contextual information.

$$\mathbf{F}i = \mathrm{Conv}d_i(\mathbf{X}), \quad i \in 1, 2, \ldots, K. \tag{17}$$

To adaptively fuse these features, CMKB generates dynamic weights using adaptive average pooling, two 1×1 convolutions, and softmax

$$\mathbf{w} = \mathrm{Softmax}(\mathbf{W}_2, \delta(\mathbf{W}_1, \alpha(\mathbf{X}))) \tag{18}$$

leading to the aggregated representation

$$\mathbf{F}\mathrm{dyn} = \sum i = w_i, \mathbf{F}_i. \tag{19}$$

This mechanism adaptively selects receptive fields at each position, balancing semantic abstraction with spatial precision and maintaining stable representations under sparse pixel distributions.

2) Local Feature Refinement: Although dynamic fusion improves scale adaptability, it may blur edges and weaken textures. CMKB, therefore, introduces a local refinement pathway

$$\mathbf{F}\mathrm{loc} = \mathrm{Conv}3 \times 3\left(\sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times 3}(\mathbf{X})\right)\right)\right) \tag{20}$$

which applies two successive convolutions to refine local spatial responses. The nongrouped convolutions reorganize local structures and strengthen boundaries crucial for small object delineation.

3) Channel Attention: A lightweight channel-attention branch adaptively reweights the fused features. The intermediate representation is

$$\mathbf{F}\mathrm{ref} = \mathbf{F}\mathrm{dyn} + \mathbf{F}\mathrm{loc} \tag{21}$$

followed by channel weighting

$$\mathbf{A}c = \sigma(\mathbf{W}4, \delta(\mathbf{W}3, \mathrm{AAP}(\mathbf{F}\mathrm{ref}))) \tag{22}$$

$$\mathbf{F}\mathrm{att} = \mathbf{F}\mathrm{ref} \odot \mathbf{A}c. \tag{23}$$

Finally, CMKB outputs

$$\mathbf{Y} = \mathrm{Conv}1 \times 1(\mathbf{F}\mathrm{att}). \tag{24}$$

By integrating the three pathways, the CMKB balances global perception and local fidelity, mitigates degradation caused by sparse representations, and enhances the detector's discriminative ability in small object scenarios.

E. Fine-Grained Small Object Detector

Small objects in RSIs occupy only a limited number of pixels and are highly susceptible to spatial information loss caused by repeated downsampling, causing fine-grained details to fade during feature propagation. Consequently, their structural boundaries and texture cues become less distinct, leading to incomplete semantic representations and unstable feature responses. This degradation increases the risk of missed detections, as small object signals may be attenuated in deeper layers, and of false positives, as background clutter with similar low-level patterns may activate the detector. Existing multiscale detection heads may struggle under these conditions because they cannot simultaneously preserve fine spatial details and sufficient semantic information, particularly in densely distributed small object regions where feature ambiguity is more pronounced.

To mitigate these limitations, we introduce FSDetector, a dedicated detection branch integrated into the YOLOv8 architecture. FSDetector is designed to retain fine-grained spatial cues and improve the feature representation of small objects by providing a higher-resolution feature level for small object prediction. By operating on high-resolution shallow features, it preserves local texture and boundary information that may otherwise be attenuated during deeper feature propagation. At the same time, the retained contextual information supports the discrimination of object responses from background patterns.

Formally, given the shallow high-resolution feature map $\mathbf{F}s \in \mathbb{R}^{B\times C_s\times H_s\times W_s}$ extracted from the backbone, the feature enhancement and detection process of FSDetector can be expressed as follows:

$$\mathbf{F}enh = \Phi_{enh}(\mathbf{F}s) \tag{25}$$

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where $\Phi enh(\cdot)$ denotes the fine-grained enhancement operation. This operation refines local structures and reinforces boundary-related responses through spatially adaptive feature recalibration. The resulting feature is then used as the input to the subsequent projection layer.

Subsequently, the enhanced feature $\mathbf{F}enh$ is passed through a convolutional projection layer to improve feature compactness and discriminability

$$\mathbf{F}proj = \text{Conv}3 \times 3\,(\mathbf{F}enh) \tag{26}$$

where $\text{Conv}_{3\times3}(\cdot)$ aggregates local contextual information while maintaining spatial localization. This operation further refines the feature representation before prediction and facilitates the separation of object and background responses.

Finally, the detection prediction is performed as follows:

$$\hat{\mathbf{Y}}s = \text{Detect}\,(\mathbf{F}proj) \tag{27}$$

where $\hat{\mathbf{Y}}_s$ represents the prediction result of the small object branch. The refined feature representation provides spatially detailed information for classification and localization, particularly for small objects with dense spatial distributions or low contrast.

Following the high-resolution detection strategy available in YOLOv8, FSDetector incorporates the $P2$ detection level to retain fine-grained spatial information, forming a four-scale detection hierarchy of $P2, P3, P4$, and $P5$. Instead of modifying the detection hierarchy itself, the FSDetector focuses on improving the quality of high-resolution features through the proposed feature enhancement pipeline, thereby facilitating the use of shallow spatial details for small object detection. The additional $P2$ prediction level allows small objects to be predicted from a higher-resolution feature map, reducing the spatial information loss associated with deeper prediction levels. The richer spatial details at this level also provide additional cues for distinguishing small objects from visually similar background patterns.

Moreover, FSDetector works together with the upstream CDAM and DPFM to form a progressive feature-processing pipeline. In particular, the CDAM alleviates semantic attenuation by reconstructing structure–detail representations, thereby retaining discriminative cues for small objects during feature propagation. DPFM balances local and global perception to reduce the dominance of large-scale contextual information and maintain responses across different object scales. Building on these refined representations, FSDetector further processes fine-grained spatial features at high resolution for prediction. This hierarchical design allows the different modules to address feature representation, multiscale perception, and high-resolution prediction at different stages of the detection pipeline.

Overall, the proposed modules form a progressive feature enhancement pipeline for remote sensing small object detection. In particular, the CDAM first alleviates semantic degradation by reconstructing structural and detail information, thereby strengthening feature representation in weak-texture regions. DPFM subsequently performs adaptive multiscale perception to balance local details and global contextual information, improving the representation of objects with different spatial extents. CMKB further refines the feature representation through collaborative multikernel perception and adaptive feature refinement, facilitating the extraction of heterogeneous object patterns. Finally, FSDetector exploits high-resolution shallow features to retain spatial details that may be lost during repeated downsampling, thereby providing a higher-resolution prediction pathway for small and densely distributed objects. Through the combined use of these modules, the CoCAPNet integrates semantic representation, multiscale perception, structural refinement, and high-resolution prediction within a unified detection pipeline.

Since FSDetector follows the existing high-resolution detection paradigm of YOLOv8, the introduced $P2$ branch does not alter the fundamental optimization strategy of the detector. Instead, it provides an additional prediction scale for exploiting the fine-grained representations generated by the preceding feature refinement modules.

IV. Experiments

This section begins by introducing publicly available datasets, each presenting unique challenges in remote sensing object detection. Next, we provide a detailed description of the evaluation metrics and experimental setup. An ablation study is conducted to validate the proposed approach and demonstrate the individual contributions of each module. The results, including detection accuracy, feature extraction performance, and comparisons with state-of-the-art methods, are discussed in the following subsections. The effectiveness of CoCAPNet is further validated by integrating it into different network architectures and datasets, particularly within the YOLOv8 [29] framework.

A. Experimental Datasets Description

1) RSOD [30]: The RSOD dataset, developed by Wuhan University, consists of various object categories, such as aircraft, oil tanks, overpasses (OPs), and playgrounds. The dataset contains 976 images and 6950 object instances: 4493 aircraft, 1586 oil tanks, 191 playgrounds, and 180 OPs. The images range in size from 512 × 512 to 1961 × 1193 pixels, with spatial resolutions between 0.3 and 3 m. The dataset is divided into training (75%) and validation (25%) sets.

2) DIOR [31]: The DIOR dataset, a benchmark for optical remote sensing object detection, consists of 23 463 images and 192 472 object instances across 20 categories. These categories include real-world objects, such as airplanes (ALs), airports (ATs), baseball fields (BFs), basketball courts (BCs), bridges (BGs), chimneys (CMs), dams (DMs), expressway service areas (EAs), toll stations (ES), vehicles (VHs), harbors (HBs), OPs, golf courses (GCs), ground track fields (GFs), ships (SPs), stadiums (SDs), storage tanks (STs), tennis courts (TCs), train stations (TSs), and windmills (WMs). The spatial resolution ranges from 0.5 to 30 m. The dataset is divided into a training set (5862 images), a validation set (5863 images), and a test set (11 738 images).

3) NWPU VHR-10 [32]: Released by Northwestern Polytechnic University in 2014, the NWPU VHR-10 dataset is a commonly used resource in remote sensing applications.

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TABLE I
TRAINING ENVIRONMENT AND HARDWARE PLATFORM PARAMETERS

Hardware	Configuration
CPU	i9-12900K
GPU	NVIDIA GeForce RTX 3090
GPU memory size	24G
Operating system	Win10
Deep learning architecture	Pytorch1.12.1 + Cuda11.3 + cudnn11.3

The most recent version contains 1172 images, classified into ten categories: airplanes, ships, STs, baseball diamonds, TCs, BCs, GFs, harbors, bridges, and vehicles. These categories are further split into positive and negative subsets. The spatial resolutions of the images range from 0.08 to 2 m.

4) AI-TOD [33]: AI-TOD is a specialized dataset designed for detecting tiny objects in aerial imagery. On the contrary to other remote sensing datasets, AI-TOD contains objects averaging 12.8 pixels in size, much smaller than those in traditional datasets. The dataset includes 28 036 aerial images and 700 621 object instances, divided into eight categories, including airplanes, bridges, and STs. The dataset follows official guidelines, including a training set of 11 214 images, a validation set of 2804 images, and a test set of 14 018 images.

B. Evaluation Metrics

The evaluation metrics used are the average precision (AP) for each class and the mean AP (mAP) across all classes, defined as follows:

$$\mathrm{AP} = \int_0^1 \mathrm{P_r}\left(R_e\right)\, dR_e \tag{28}$$

$$\mathrm{mAP} = \frac{1}{\mathrm{N_{cls}}} \sum_{i=1}^{\mathrm{N_{cls}}} \mathrm{AP}_i \tag{29}$$

where P_r and R_e denote the precision and the recall, respectively, and N_{cls} serves as the total number of classes. The precision P_r and the recall R_e can be expressed as follows:

$$\mathrm{P_r} = \frac{\mathrm{N_{TP}}}{\mathrm{N_{TP}} + \mathrm{N_{FP}}} \tag{30}$$

$$\mathrm{R_e} = \frac{\mathrm{N_{TP}}}{\mathrm{N_{TP}} + \mathrm{N_{FN}}} \tag{31}$$

where N_{TP}, N_{FP}, and N_{FN} denote the amount of true positives (TP_s), false positives (FP_s), and false negatives (FN_s), respectively. In particular, a TP refers to data instances that are both correctly identified and accurately classified. Conversely, an FP denotes instances that are incorrectly detected or misclassified. In addition, FN describes instances that fail to be detected. A prediction qualifies as a TP if its intersection over union (IoU) with the ground truth exceeds 0.5 during inference; otherwise, it is classified as an FP.

C. Implementation Details

The experiments were conducted in a PyTorch 1.12.1 environment on a Windows ten system, using an NVIDIA RTX 3090 GPU with 24 GB of RAM, as detailed in Table I. Detection performance was evaluated using the mAP at an IoU threshold of 0.5. The hardware and environment configurations used during training are detailed in Table II.

TABLE II
KEY PARAMETERS SET DURING MODEL TRAINING

Parameters	Setup
epochs	600
Momentum	0.932
Initial learning rate	0.01
Final learning rate	0.01
Weight decay	0.0005
Batch size	4
Input image size	640×640
Optimizer	SGD

D. Ablation Study and Analysis

We evaluate the contribution of each component in CoCAPNet by systematically integrating the CDAM, DPFM, and the CMKB into the baseline model and assessing their impact on performance. The ablation study was carried out using the RSOD dataset. Table III shows how the inclusion or exclusion of each module impacts the evaluation metrics. For a more in-depth analysis, we visualize CoCAPNet's detection results on the RSOD dataset, as shown in Fig. 6, highlighting outcomes for four representative categories.

The ablation results show that the four modules contribute differently to the detection performance. For airplanes with substantial directional variation, CoCAPNet accurately detects targets from multiple orientations, confirming its adaptability in complex scenarios. In densely arranged oil tank areas, the model effectively resolves feature confusion, ensuring reliable detection in high-density scenes. These findings further highlight the superiority of CoCAPNet for multiscale object detection in challenging RSIs.

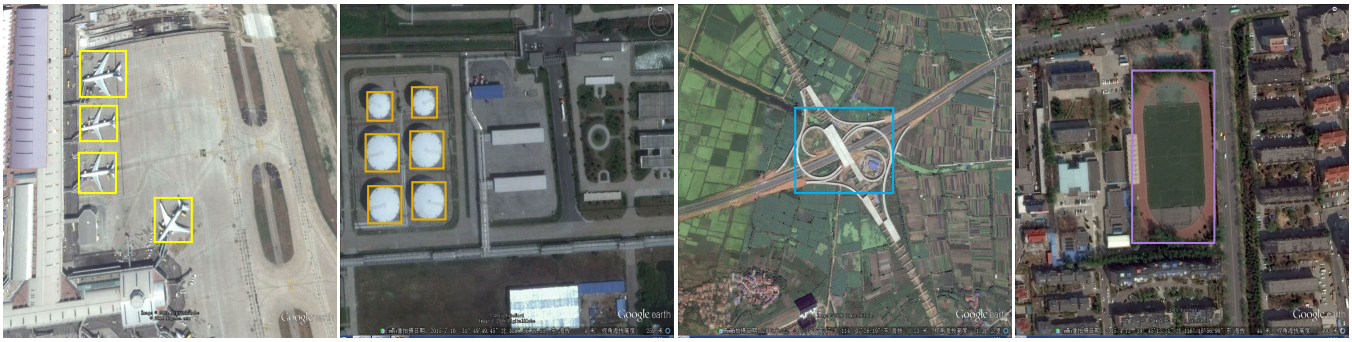
1) Effect of CDAM: As shown in the second row of Table III, applying CDAM leads to a clear performance gain. When used alone, mAP_{50} increases from 89.9% to 92.0%. This indicates that CDAM improves the representation of object-related cues in the RSOD scenes.

2) Effect of DPFM: The third row of Table III shows that integrating DPFM improves mAP_{50} to 93.1% and recall to 88.5%. The DPFM enables dynamic weighting of spatial regions across scales, strengthening spatial alignment and improving the discrimination of critical regions.

3) Effect of CMKB: In the fourth row of Table III, adding CMKB results in an increase of mAP_{50} to 92.3%, with recall reaching 82.2%. CMKB helps the model better emphasize structurally relevant regions and enhances the detection of small or overlapping objects.

4) Effect of FSDetector: When used alone, the FSDetector increases mAP50 from 89.9% to 93.5%, indicating the benefit of introducing the high-resolution $P2$ prediction level. When combined with CDAM, DPFM, and the CMKB, the full model further reaches 94.5% mAP50 and 68.7% $\mathrm{mAP}_{50:95}$. These results verify their complementary contributions, collectively reducing background noise, alleviating feature confusion, and reinforcing attention to critical spatial regions.

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(a) (b) (c) (d)

Fig. 6. Visualizing the detection results of our CoCAPNet on the RSOD dataset. (a) Airplane. (b) Oil tank. (c) Overpass. (d) Playground.

TABLE III

ABLATION EXPERIMENTS FOR CDAM, DPFM, CMKB, AND FSDETECTOR ON THE RSOD DATASET

CDAM	DPFM	CMKB	FSDetector	$mAP_{50}(\%)$	$mAP_{50:95}(\%)$	Recall(%)
				89.9	64.0	91.6
✓				92.0	61.2	86.6
	✓			93.1	66.4	88.5
		✓		92.3	65.3	82.2
			✓	93.5	65.2	90.4
✓	✓			93.1	63.1	89.4
✓			✓	92.7	62.5	90.0
✓	✓	✓		94.0	67.9	**91.7**
✓	✓	✓	✓	**94.5**	**68.7**	91.5

TABLE IV

PERFORMANCE COMPARISONS ON THE DIOR DATASET. BEST RESULTS ARE IN BOLD. *RESULTS FROM [48]

Method	mAP(%)	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM
Faster R-CNN [34]	66.9	81.1	76.9	89.8	80.5	45.1	81.7	63.4	80.9	63.2	73.3	78.3	43.3	58.4	68.5	89.1	59.0	81.1	60.2	47.8	79.3
CF2PN [35]	67.3	78.3	78.3	76.5	88.4	37.0	71.0	59.9	71.2	51.2	75.6	77.1	56.8	58.7	76.1	70.6	55.5	88.8	50.8	36.9	86.4
Cascade R-CNN [36]	67.4	81.2	81.4	90.1	81.1	46.9	81.5	68.1	84.2	65.3	74.5	81.6	37.8	60.4	68.9	88.8	60.5	81.0	57.5	47.7	80.7
CSFFNet [37]	68.0	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9
SRAF-Net [38]	69.7	88.4	76.5	92.6	87.9	35.8	83.8	58.6	86.8	66.8	76.4	82.8	16.2	58.0	59.4	80.9	55.6	90.6	52.0	53.2	91.0
ASSD [39]	71.1	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3
RSADet [40]	72.2	73.6	86.0	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5
SB-MSN [41]	70.3	79.6	82.2	76.4	89.8	45.6	78.2	64.8	58.9	59.3	79.2	82.4	51.8	60.8	74.4	79.7	66.4	85.6	65.4	55.3	86.5
TMAFNet [42]	72.9	92.2	77.7	75.0	91.3	47.1	78.6	53.6	67.1	66.2	78.5	76.3	64.9	61.4	90.5	72.2	75.4	90.7	62.1	55.2	83.2
AGMF-Net [43]	73.2	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	65.0	61.7	91.7	78.6	75.8	90.7	60.0	58.0	83.1
LGDA [44]	74.1	79.6	83.4	81.8	89.2	52.8	73.7	68.1	84.5	61.2	77.3	81.9	60.1	55.2	89.7	64.9	82.7	85.2	63.9	58.5	88.6
CoF-Net [49]	75.8	84.0	85.3	82.6	90.0	47.1	80.7	73.3	89.3	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2
GLSANet [45]	77.9	95.8	78.9	92.9	87.9	50.7	81.1	55.5	79.8	74.1	71.6	87.6	66.4	65.5	95.2	92.4	86.3	94.8	50.6	62.1	89.2
AFSNet [46]	78.9	91.6	83.4	**96.1**	**94.9**	56.2	87.6	**79.4**	64.6	76.8	**87.9**	73.3	60.5	61.7	94.3	87.4	77.0	89.2	65.7	57.9	92.9
CIOD [47]*	69.5	80.3	74.3	71.7	89.1	43.9	78.6	60.2	60.3	62.6	75.4	74.6	61.8	57.3	89.5	54.2	74.5	85.4	57.7	56.1	81.6
GOCPK [48]*	70.2	75.6	81.1	71.9	86.8	43.8	79.4	74.0	59.6	58.4	80.0	76.2	60.2	58.0	83.2	60.0	70.5	87.1	63.1	54.7	80.1
CoCAPNet (ours)	**86.3**	**97.4**	**90.7**	94.5	90.8	**58.5**	**87.9**	73.7	96.7	88.0	88.1	**90.6**	**76.1**	**76.7**	**96.2**	96.5	**89.8**	**97.0**	67.4	73.3	**96.4**

To further examine these effects, Fig. 7 visualizes feature maps for airplanes, oil tanks, ships, and vehicles. As seen in Fig. 7(b), raw features contain notable background interference, such as sea textures and wave patterns, and the features of oil tanks remain heavily entangled. With CDAM applied, Fig. 7(c) shows clear suppression of irrelevant background responses and stronger emphasis on object structures. Adding DPFM, as shown in Fig. 7(d), further enhances feature separation in densely distributed scenarios. Fig. 7(e) illustrates that incorporating the CMKB sharpens object boundaries and improves attention consistency in cluttered regions. Finally, Fig. 7(f) demonstrates that FSDetector restores finer spatial cues and improves the recognition of small, dense, or partially occluded objects.

E. Comparison Experiments and Analysis

This section presents a comparative analysis of CoCAPNet and other state-of-the-art detectors, evaluated on the DIOR, NWPU VHR-10, and AI-TOD datasets. The corresponding results are summarized in Tables IV–VI.

1) Results on DIOR: We evaluate CoCAPNet on the DIOR dataset, as shown in Table IV, and compare it with several advanced detectors, including Faster R-CNN [34], CF2PN [35], Cascade R-CNN [36], CSFFNet [37], SRAFNet [38], ASSD [39], RSADet [40], SB-MSN [41], TMAFNet [42], AGMFNet [43], LGDA [44], CoF-Net [49], GLSANet [45], AFSNet [46], as well as two recently proposed Transformer-based methods, CIOD [47] and GOCPK [48]. As reported in

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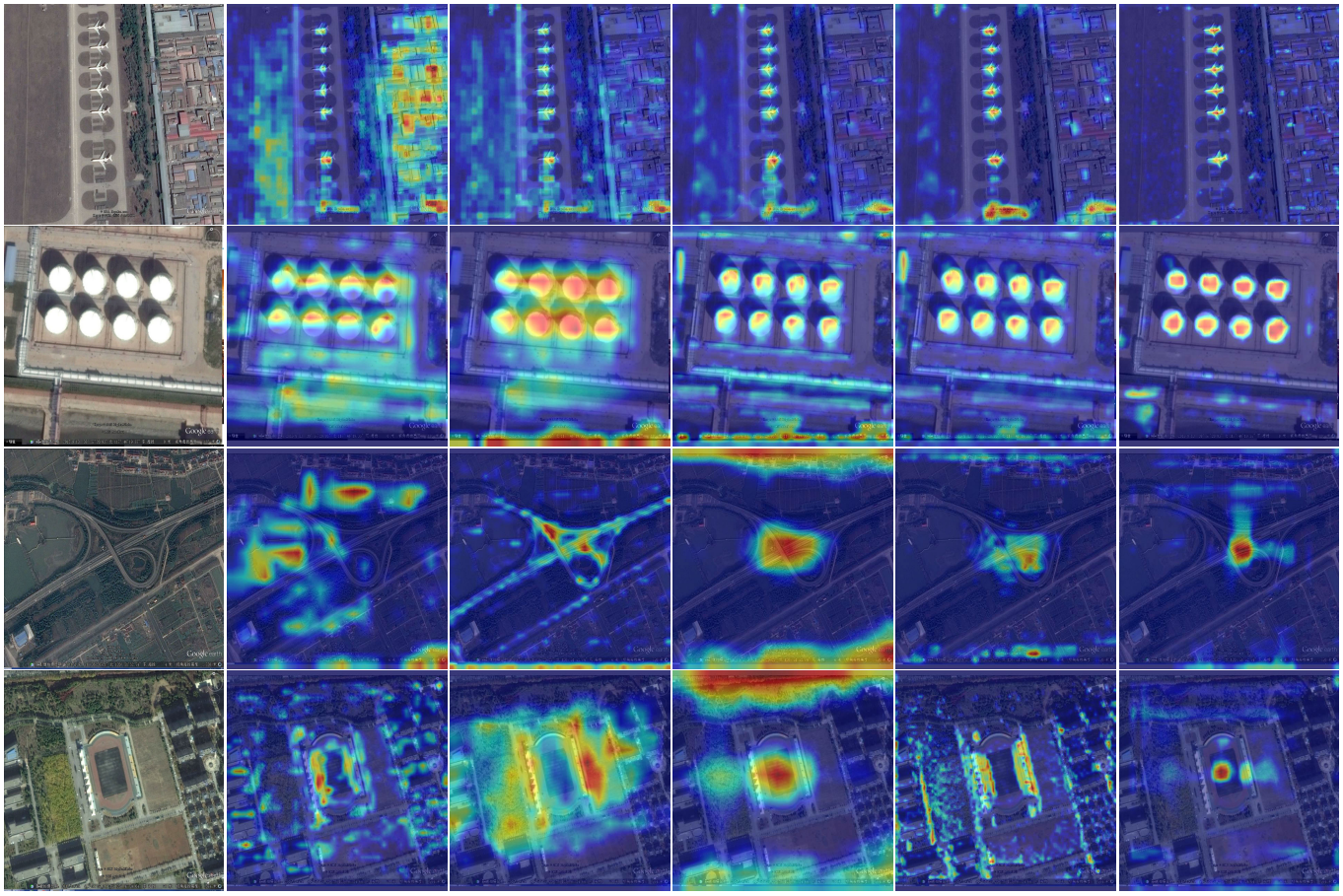


Fig. 7. Influence of CDAM, DPFM, CMKB, and FSDetector on feature extraction. We selected four challenging RSIs to validate the proposed modules, which respectively represent orientation changes, scale variations, weak texture, and densely arranged small objects. The first row illustrates airplanes, which exhibit substantial variations in orientation among objects. The second row displays oil tanks, which are characterized by their dense arrangement as small objects. The third and fourth rows show ships and vehicles, respectively. Ships are influenced by the textured background of ocean waves, whereas vehicles encounter a combination of all these challenges.

TABLE V

PERFORMANCE COMPARISONS ON THE NWPU VHR-10 DATASET. THE BEST RESULTS ARE MARKED IN BOLD. *RESULTS FROM [55]; †FROM [57]

Method	mAP	Airplane	Ship	ST	BD	TC	BC	GT	Harbor	Bridge	Vehicle
HRBM [50]	87.1	99.7	90.8	90.6	92.9	90.3	80.1	90.8	80.3	68.5	87.1
YOLOv3 [51]	87.3	99.6	81.3	80.3	98.3	80.6	81.8	99.5	74.1	89.6	87.0
Faster R-CNN [34]	88.3	**100**	85.3	**100**	95.9	87.6	92.1	99.7	92.1	43.4	80.6
FPN [16]	90.8	**100**	90.9	**100**	96.8	90.7	95.1	**100**	93.7	50.9	90.1
EGAT [52]	92.0	97.3	90.9	97.2	96.5	86.6	94.5	94.2	86.2	80.1	90.8
YOLOv7 [53]	92.9	99.5	**96.7**	96.6	97.8	93.9	90.8	95.6	89.5	83.7	89.4
YOLOv8 [29]	93.6	99.9	92.6	96.7	97.9	93.9	90.8	96.8	90.7	84.7	90.6
LGDA [44]	94.1	98.8	91.3	95.7	98.4	96.9	94.6	94.2	93.1	89.8	88.0
CoF-Net [49]	94.5	**100.0**	90.6	96.4	98.9	91.5	95.7	99.7	92.1	89.6	90.6
GLSANet [45]	94.6	99.9	94.0	97.1	**99.4**	98.8	86.1	99.5	97.5	84.8	86.8
Swin Transformer [54]*	87.7	93.3	86.7	87.3	86.6	82.7	81.9	90.5	91.4	89.9	87.1
Compact DETR [55]*	93.7	98.9	89.5	90.9	98.9	**99.6**	99.2	**100.0**	86.9	81.2	91.9
SED-DETR [56]†	94.4	99.9	91.3	97.7	94.3	94.5	93.7	94.6	94.6	**93.6**	90.1
FEMS-DETR [57]†	95.0	99.5	92.5	93.6	99.0	98.4	97.3	99.5	**97.9**	84.0	88.0
CoCAPNet (ours)	**95.8**	99.5	93.6	99.4	97.1	98.0	**99.5**	99.3	91.8	82.4	**97.5**

Table IV, CoCAPNet achieves the highest mAP of 86.3%, surpassing the second-best approach AFSNet (78.9%) by 7.4 percentage points and demonstrating a substantial performance gain on DIOR. Among the Transformer-based baselines, GOCPK (70.2%) and CIOD (69.5%) rank in the mid-to-lower tier, reflecting the inherent difficulty of

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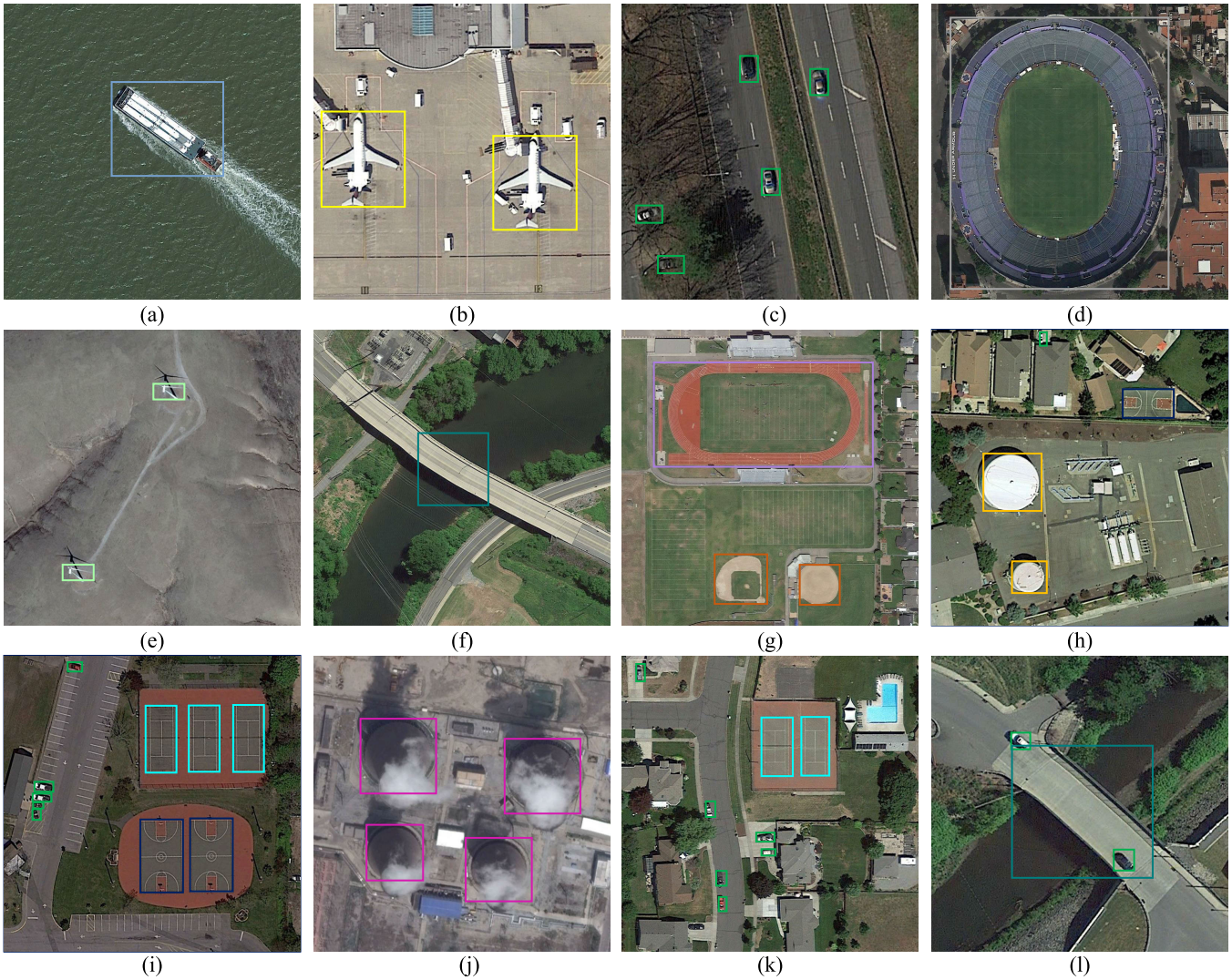


Fig. 8. Detection results of CoCAPNet in DIOR for representative scenarios. (a) Ship. (b) Airplane. (c) Vehicle. (d) Stadium. (e) WM. (f) Bridge. (g) GF and BF. (h) Oil tank and vehicle. (i) Vehicle, TC and BC. (j) Chimney. (k) TC and vehicle. (l) Bridge and vehicle.

TABLE VI

PERFORMANCE COMPARISONS ON THE AI-TOD DATASET. BEST RESULTS ARE MARKED IN BOLD. *RESULTS ARE FROM [66]; ‡ RESULTS ARE FROM THE ORIGINAL PAPER

Methods	$\mathbf{mAP}_{50}$	$\mathbf{mAP}_{50:95}$
QueryDet [58]*	0.293	0.122
DN-DETR [59]*	0.326	0.128
RT-DETR [60]*	0.417	0.180
H-Deformable-DETR [61]*	0.366	0.139
M-CenterNet [33]	0.407	0.145
Cascade R-CNN [36]	0.308	0.138
DetectoRS [62]	0.328	0.148
DetectoRS + NWD [63]	0.493	0.208
SP-YOLOv8s [64]	0.483	0.227
HANet [65]	0.537	0.221
L-FFCA-YOLO [12] ‡	0.583	0.255
CoCAPNet (ours)	**0.591**	**0.256**

DIOR with its 20 diverse categories. Both show selective strengths—GOCPK achieves 87.1% on TC and 83.2% on SP via geometric prior modeling, while CIOD attains 89.5% on SP and 85.4% on TC through capsule inference. However, they struggle with high-variability categories such as BG (43.8%/43.9%) and HB (60.2%/61.8%), where contextual priors are insufficient to resolve complex background ambiguity. These results further underscore the necessity of the collaborative multiscale perception and context-affine modeling strategy employed by CoCAPNet. Categorywise, CoCAPNet ranks first in 11 of the 20 classes. Notably, in categories such as BG and ES, its precision improves upon the second-best method by 2.3% and 16.7%, respectively. Even in challenging categories like BG, where precision remains relatively low (58.5%), CoCAPNet still outperforms AFSNet by 2.3%. The integration of FSDetector further enhances the ability to focus on small and overlapping objects, contributing to the overall accuracy improvements.

Fig. 8 illustrates the detection results of CoCAPNet across representative DIOR scenarios. Despite complex and cluttered backgrounds, the model effectively identifies objects such

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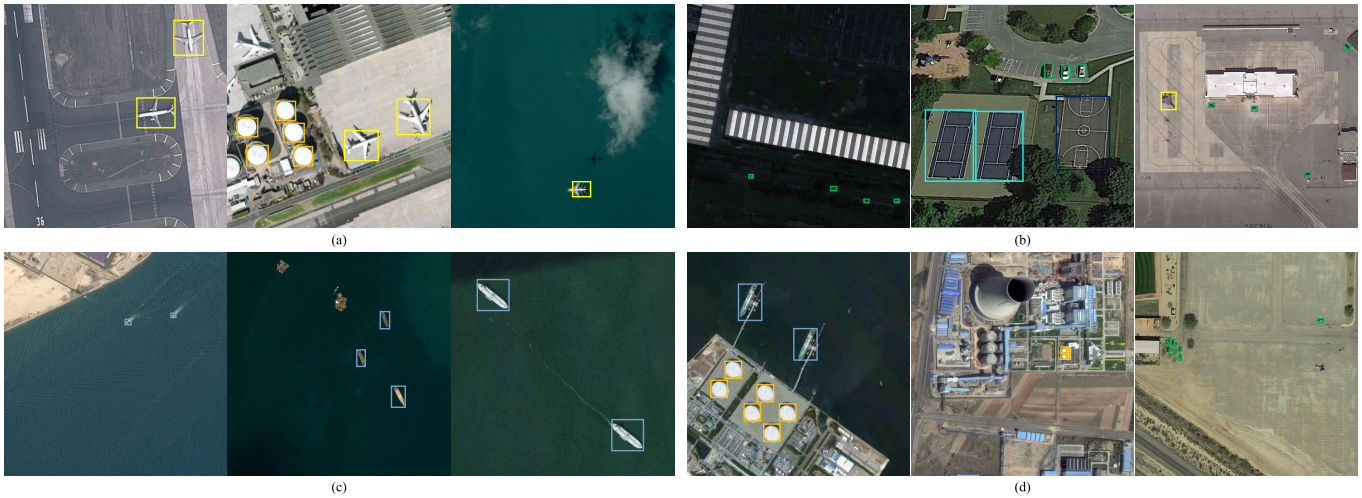


Fig. 9. Visualizing the detection results of our CoCAPNet on the NWPU VHR-10 and AI-TOD. (a) Airplane and oil tank. (b) Vehicle, TC, and BC. (c) Ship. (d) Vehicle, ship, and oil tank.

as the WM in Fig. 8(e) and the oil tank in (h). These visual outcomes highlight the effectiveness of the CDAM in alleviating weak textures and restoring attenuated semantics, enabling more reliable extraction of contextual cues. The results collectively verify that CDAM, DPFM, CMKB, and the FSDetector work synergistically to deliver robust multiscale perception in challenging RSI environments.

2) Results on NWPU VHR-10: Table V compares CoCAPNet with state-of-the-art detectors on the NWPU VHR-10 dataset, including HRBM [50], YOLOv3 [51], faster R-CNN [34], FPN [16], EGAT [52], YOLOv7 [53], YOLOv8 [29], LGDA [44], CoF-Net [49], GLSANet [45], and four recently proposed Transformer-based methods: Swin Transformer V2 [54], compact DETR [55], SED-DETR [56], and FEMS-DETR [57]. CoCAPNet achieves the highest mAP of 95.8%, surpassing the second-best method FEMS-DETR (95.0%) by 0.8 percentage points. Among the Transformer-based baselines, FEMS-DETR and SED-DETR deliver competitive performance, benefiting from frequency-enhanced and scale-enhanced attention mechanisms, respectively; compact DETR achieves 93.7% mAP as a lightweight real-time detector; while Swin Transformer V2 yields a relatively modest 87.7%, suggesting that generic vision backbones, without task-specific adaptations, are suboptimal for remote sensing small object detection. Categorywise, CoCAPNet attains superior performance on challenging classes such as Vehicle (97.5%), substantially outperforming compact DETR (91.9%) and Swin Transformer V2 (87.1%). These results demonstrate that the synergistic collaboration of the CDAM, DPFM, CMKB, and the FSDetector effectively mitigates semantic degradation and preserves fine-grained spatial details, leading to robust detection performance on the NWPU VHR-10 benchmark.

3) Results on AI-TOD: Table VI presents a comparison between CoCAPNet and other state-of-the-art detectors on the AI-TOD dataset. The competing methods include QueryDet [58], DN-DETR [59], M-CenterNet [33], Cascade R-CNN [36], DetectoRs [62], DerectoRS + NWD [63], SP-YOLOv8s [64], HANet [65], and L-FFCA-YOLO [12]. On this dataset, CoCAPNet achieves the highest performance with an mAP_{50} of 59.1% and an $mAP_{50:95}$ of 25.6%. This outperforms the second-best method, L-FFCA-YOLO, by 0.8% in mAP_{50} and 0.1% in $mAP_{50:95}$. The use of the FSDetector, combined with multiscale feature extraction, enhances CoCAPNet's ability to handle dense small object arrangements and significant scale variations. These advances allow CoCAPNet to effectively address challenges such as complex backgrounds and low-contrast objects, further highlighting its robustness. Qualitative detection results of CoCAPNet on the NWPU VHR-10 and AI-TOD datasets are presented in Fig. 9.

TABLE VII

GENERALITY EXPERIMENTS OF CDAM, DPFM, AND CMKB ON RSOD AND DIOR

Model	Size	Dataset	Baseline	CDAM	DPFM	CMKB
YOLOv9	c	DIOR	79.1	**79.5**	**79.8**	**81.5**
YOLOv9	s	DIOR	72.5	**73.2**	**73.4**	**72.7**
YOLOv9	s	RSOD	90.0	**90.4**	89.5	**91.9**
YOLOv10	s	RSOD	92.5	**92.9**	**92.9**	**93.2**
YOLOv10	n	RSOD	91.2	**92.0**	**91.9**	**92.6**
YOLOv10	l	RSOD	92.0	**93.6**	**92.3**	**92.4**
YOLOv10	m	DIOR	85.5	**86.0**	**85.7**	**86.1**

Note: Bold indicates improvement over Baseline.

F. Generality Experiment

The effectiveness and transferability of the CDAM, DPFM, and CMKB were further validated across different YOLO architectures and datasets. As reported in Table VII, which consolidates the results of all three modules under identical configurations, each module consistently enhances detection performance across most settings, demonstrating strong generalization capability.

Among the three modules, the CMKB provides the largest improvement in several configurations, although its gain varies across architectures and datasets. Notably, on the YOLOv9-c [67]/DIOR configuration, the CMKB yields a substantial gain of 2.4% (from 79.1% to 81.5%), and on

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YOLOv9-s [67]/RSOD, it achieves a 1.9% improvement (from 90.0% to 91.9%). These results highlight the effectiveness of CMKB's multikernel perception and channel-aware refinement in enhancing feature discriminability across diverse scenarios.

CDAM also delivers consistent performance gains across all tested configurations. In particular, it achieves a notable 1.6% improvement on YOLOv10-l [68]/RSOD (from 92.0% to 93.6%), and demonstrates robust adaptability on both the YOLOv9 and YOLOv10 architectures across different model scales, underscoring its plug-and-play versatility.

DPFM yields moderate yet steady improvements in most cases. However, a slight performance degradation is observed on the YOLOv9-s/RSOD configuration (from 90.0% to 89.5%). This minor exception may be attributed to the inherent tradeoff between local detail preservation and global receptive field expansion when applied to relatively shallow architectures on small-scale datasets.

Overall, these results collectively verify that CDAM, DPFM, and the CMKB possess strong generalization ability and can effectively improve small object detection performance across different YOLO architectures and dataset scenarios, with CMKB demonstrating the most substantial and consistent effectiveness.

V. Conclusion

This article presents CoCAPNet, a small object detection framework for RSIs. The network alleviates semantic degradation, feature suppression, and insufficient structural expression through four complementary modules. CDAM separates structure- and detail-dominant responses through spatial-domain decomposition and adaptive fusion. DPFM combines responses from multiple receptive-field ranges to balance local details and contextual information. CMKB combines dilated depthwise convolutions, local refinement, and channel reweighting to refine multiscale feature responses. The FSDetector operates on high-resolution shallow features to recover spatial details lost during downsampling, thereby improving the localization of small objects.

Experiments on the DIOR, NWPU VHR-10, and the AI-TOD show that CoCAPNet achieves mAP values of 86.3%, 95.8%, and 59.1%, respectively. Generalization studies further show that CDAM, DPFM, and the CMKB transfer effectively across different detection frameworks, underscoring their plug-and-play versatility. Overall, the proposed framework provides an efficient multidimensional feature enhancement solution for RSIs. Although integrating multiple adaptive modules increases computational cost and optimization complexity, future work will focus on improving efficiency and strengthening robustness for real-world deployment.

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